Let $f(x) \in F[x]$ be given with the irreducible factorization

$$f(x) = f_1(x) \cdots f_n(x).$$

Since every Automorphism fixing F maps root of f_i to the root of f_i, the Same irreducible Polynomial, therefore, for the Splitting field K of f,

$$\mathrm{Aut}(K/F) \hookrightarrow S_{f_1} \times \cdots \times S_{f_n}.$$

With such an observation, in this paper, we figure out concretely the Galois group of the Polynomials.

Def. Let $x_1, \cdots, x_n$ be indeterminates.
The elementary symmetric functions $S_1, S_2, \cdots, S_n$ are defined by

$$S_1 = x_1 + x_2 + \cdots + x_n$$
$$S_2 = x_1x_2 + x_1x_3 + \cdots + x_1x_n + x_2x_3 + x_2x_4 + \cdots + x_{n-1}x_n$$
$$\vdots$$
$$S_n = x_1x_2 \cdots x_n$$

And, the general polynomial of degree n is the polynomial

$$f(x) = (x - x_1)(x - x_2) \cdots (x - x_n).$$

Note that

$$f(x) = x^n - S_1x^{n-1} + S_2x^{n-2} - \cdots + (-1)^n \cdot S_n$$

Thm. Let $x_1, \cdots, x_n$ be indeterminates of a field F, and $S_1, \cdots, S_n$ be symmetric functions. Then, $F(x_1, \cdots, x_n)$ is Galois extension of $F(S_1, \cdots, S_n)$ with

$$\mathrm{Gal}\left(F(x_1, \cdots, x_n) / F(S_1, \cdots, S_n)\right) \cong S_n.$$

Proof. Since $F(x_1, \cdots, x_n)$ is the splitting field of the polynomial

$$f(x) = (x - x_1) \cdots (x - x_n) = x^n - S_1x^{n-1} + \cdots + (-1)^n \cdot S_n$$

over $F(S_1, \cdots, S_n)$ and f is separable, so the first assertion is obtained.
And, since degree of f is n, so $[F(x_1, \cdots, x_n) : F(S_1, \cdots, S_n)] \le n!$.
Meanwhile, for any permutation $\sigma \in S_n$, a induced map

$$\bar{\sigma} : F(x_1, \cdots, x_n) \to F(x_1, \cdots, x_n) : \frac{g(x_1, \cdots, x_n)}{f(x_1, \cdots, x_n)} \mapsto \frac{g(x_{\sigma(1)}, \cdots, x_{\sigma(n)})}{f(x_{\sigma(1)}, \cdots, x_{\sigma(n)})}$$

is an Automorphism which fixes $F(S_1, \cdots, S_n)$, so

$$S_n \hookrightarrow \mathrm{Aut}\left(F(x_1, \cdots, x_n) / F(S_1, \cdots, S_n)\right)$$

Since $|S_n| = n!$, so proved. ▣

「Determinant.

Define the discriminant D of $x_1, \cdots, x_n$ by the formula

$$D = \prod_{1 \leq i < j \leq n} (x_i - x_j)^2 \in \mathbb{Z}[x_1, \cdots, x_n]$$

Define the discriminant of a polynomial to be the disriminant of the roots.
Using the discriminant D, denote that

$$\sqrt{D} := \prod_{1 \leq i < j \leq n} (x_i - x_j) \in \mathbb{Z}[x_1, \cdots, x_n].$$

Note: the discriminant D is a symmetric function in $x_1, \cdots, x_n$.
And, for a permutation $\sigma \in S_n (= Sym(x_1, \cdots, x_n))$

$\sigma \in A_n$ if and only if σ preserves the sign of $\sqrt{D}$, that is, $\sigma(\sqrt{D}) = \sqrt{D}$.

Now, we obtain the following result:

「Thm. Let F be a field which is not characteristic 2.
Then, for indeterminates $x_1, \cdots, x_n$,

$$\underbrace{F(s_1, \cdots, s_n) \leq F(s_1, \cdots, s_n)(\sqrt{D})}_{\deg 2} = \underbrace{F(x_1, \cdots, x_n)^{A_n} \leq F(x_1, \cdots, x_n)}_{\deg n!/2}.$$

Moreover, from the language of the polynomial, for

$Gal(K/F) \cong A_n$ if and only if $\sqrt{D}$ is fixed by all elements in $Gal(K/F)$.

「Polynomials of degree 2.

Consider the Polynomial x^2+ax+b with root x_1 and x_2.

The discriminant is

$$D=(x_1-x_2)^2=(x_1+x_2)^2-4x_1x_2=s_1^2-4s_2=(-a)^2-4(b)=a^2-4b.$$

The polynomial is separable if and only if $x_1=x_2$ if and only if $a^2-4b\neq 0$.

Since K, the splitting field of x^2+ax+b over $\mathbb{Q}$, $\mathrm{Gal}(K/\mathbb{Q})\hookrightarrow S_2$ and

$\mathrm{Gal}(K/\mathbb{Q})$ is trivial if and only if $\sqrt{a^2-4b}\in\mathbb{Q}$.

$\mathrm{Gal}(K/\mathbb{Q})\cong C_2$ if and only if $\sqrt{a^2-4b}\notin\mathbb{Q}$

Polynomial of degree 3, Cubic.

If the cubic polynomial is reducible, then it splits either into three linear factors or a linear factor × an irreducible quadratic.

In the first case, the Galois group is trivial, and in the second case, the Galois group is C_2.

Assume that the polynomial f is irreducible.

Let K/F be the splitting field of the cubic.

Since $[K:F] \geq \deg f = 3$, so that there are only two possibilities,

$$\mathrm{Gal}(K/F) \cong A_3 \quad \text{or} \quad S_3.$$

Using the knowledge of the discriminant,

$\mathrm{Gal}(K/F) \cong A_3$ if and only if $\sqrt{D} \in F$.